2026 Competition COLLEGIATE

Claremont CA #6317 ACCEPTED

United States (Claremont) · North America · Higher Education

Edit

INSTITUTIONS

Keck Graduate Institute ↗
Keck Graduate Institute
United States (Claremont) · Riggs

Claremont McKenna College ↗

Safety Forms * Required

A collection of forms documenting how you are addressing biosafety and biosecurity in your project.

Deadlines Jul 1 · Sep 2 · Sep 23 · Oct 7 Your submission status In Progress

The purpose of this form is for everyone, including iGEM, to know you are doing your project safely and securely. This is how we know that eve assessing and managing any risks to iGEM teams, their colleagues, participants in their research, their broader communities, or the environmer you are handling these risks, we don't know. Being transparent about possible risks and how we are managing them is a key component of beir or engineer, so please answer the questions as fully as possible!

Your iGEM Safety Forms must be submitted in English to facilitate review and record-keeping.

To meet the preliminary deadline, mark this form as ready for review. This means that your preliminary version has been submitted for feedback. You may conti form after marking it ready; every time you save, it will remain marked as ready for review.

Form Review

Rules & Policies > About Your Lab > 3 About Your Project > 4 Identifying Risks > 5 Future Risks > 6 Managing Risks

STEP #3

About Your Project

Tell us about your project, the organisms and parts you'll use, your experiments, and any chemicals or AI tools involved.

Other Safety

Although all teams a Project Safety form, need to get additiona **doing their work**. In

1 Check In Form

Project Goal

6. Describe the goal of your project: what is your engineered organism (or other synthetic biology product, system, or tool) supposed to do? *

Even though your project might change, please describe the main project idea you are working on right now, including specific technical details. See the example answers for help.

Bad example answers (not enough detail):

- We are engineering *E. coli* to cure liver cancer.
- Climate change is a very important problem. Our algae will reduce CO2 emissions and fight climate change.

Good example answers:

- Our bacteria will be engineered to interact with human cells. They will detect tumor cells that express biomarkers for liver cancer. They will use invasin to enter the tumor cells, and then secrete apoptin to kill the tumor cells.
- Our algae will receive gases high in CO2. We will increase their expression of Photosystem II proteins to make them absorb more CO2 from the gas.

Organisms

7. Will you be engineering whole organisms, including viruses and cell lines in your project? *

- ◯ Yes
- ◯ No

Please explain in more detail how your project will be conducted without engineering an organism. *

8. Will you use any other organisms in your project? List organisms, including genus, species, and strain. *

For example, without engineering an organism directly, you might plan to extract RNA or DNA from the organism, or test your product on it.

Parts & Synthesis

The team should sub
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2. Animal Use Form
The team should sub
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The content
public

In order to reassure
iGEM's synthetic bio
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openly available. Any
the materials your te

Questions?

If you have any ques
form or iGEM's safet

→ **safety@igem.org**

9. As part of your project, are you planning to make / have made new parts or substantively changed existing parts in the Registry? *

- ◯ Yes
- ◯ No, our project will only use genetic parts that are already in the registry

10. Could any of your parts be hazardous on their own and/or in the context of your project? *

- ◯ Yes
- ◯ No

11. What synthesis screening is done for the DNA/RNA that your team will use? *

One set of standards for synthesis screening is maintained by the International Gene Synthesis Consortium ↗.

- ◯ We will not use any synthesized DNA/RNA
- ◯ We will order our DNA/RNA from a company, member of the IGSC
- ◯ We will order our DNA/RNA from a company, but the company is not a member of the IGSC
- ◯ We will synthesize our DNA/RNA on our own synthesis machine, which is manufactured by a member of the IGSC
- ◯ We will synthesize our DNA/RNA on our own synthesis machine, but it does not do IGSC-compliant screening
- ◯ Other (please specify)

Experiments & Chemicals

12. What experiments will you do with your organisms and parts? *

Please explain briefly. We are particularly keen to understand the boundaries or scope of your project. You should include the names of species / cell lines / strains. You should include experiments involving parts taken from other organisms, even if they are being synthesized rather than isolated from nature - you need not include any parts already in the registry.

Example answers

- Our bacteria is meant to live on plant leaves, so we will test them on tobacco (*Nicotiana benthamiana*) in a lab greenhouse.
- We want to use a protein from ants, but its sequence is unknown. So we will capture ants (*Camponotus spp.*) to extract DNA and RNA to find the sequence of the protein we want.
- Our bacteria need to interact with human cells for a medical application. We will test them in human cell culture using the HEK293 cell line.
- We are interested in a RNA-binding protein expressed in *Kluyveromyces lactis*. We have found the sequence in a paper and will have it synthesized by a commercial provider.
- Our project will not involve experiments with organisms or parts. We will run digital directed evolution experiments to identify a candidate receptor binding protein for our fungicide.

13. Are you using some kind of hazardous chemicals in your project? *

Microbiology labs often use many types of chemical reagents. For example, ethanol used to clean lab benches is highly flammable. Your team should have access to the safety data sheets for these reagents to understand the chemical hazards in your lab.

- ◯ Yes
- ◯ No

AI Tools

14. Is your team using or planning to use an Artificial Intelligence (AI) tool(s) for the development of your project? *

- ◉ Yes
- ◯ No

Which category of AI tool(s) are you using or planning to use for your project? *

Check all that apply

- [ ] Large Language Models (e.g. Claude, ChatGPT, DeepSeek, Gemini, Qwen, etc.)
- [ ] Generative AI for images or video
- [ ] Genetic Prediction Tools
- [ ] Genetic Design Tools
- [ ] Protein Structure Prediction Tools
- [ ] Protein Design Tools
- [ ] Experimental Design
- [ ] Lab Automation
- [ ] Data Analysis
- [ ] Other

What risks, if any, do you foresee arising from the use of these AI tools in your project? What measures will you take to identify and mitigate these risks? *

Complete all required fields to continue.

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iGEM Foundation · 501(c)(3) not-for-profit organization · 45 Prospect Street, Cambridge MA, 02139 USA
iGEM Europe · **iGEM HQ** · 10 rue du Mail, 75002 Paris, France

Contact